MIL-F-60336(MU)
3 MAY 1965
SUPERSEDING
(SEE 6.5)

MILITARY SPECIFICATION

FUZE, POINT DETONATING M521 METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for a fuze designated as Fuze, Point Detonating, M521 for use with 4.2 Inch Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 — Ammunition and Special Weapons, General Specification for

MIL–P–10025 — Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for

MIL–I–45607 — Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes (ABC–STD–105)

MIL–STD–109 — Quality Assurance Terms and Definitions

MIL–STD–1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

7549044 — Head, Metal Parts Assembly

7549055 — Screw, Detonator Retaining

7549056 — Screw Detonator Retaining

7549060 — Tube, Flash

7549113 — Body Assembly

FSC 1390

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7549114 — Bottom Closing Screw Assembly

7549126 — Ogive

PUBLICATIONS

ORDNANCE CORPS

IEL-7153449 — Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (dwg.) 7549044, 7549055, 7549056, 7549060, 7549113, 7549114 and 7549126 as applicable, and with all requirements specified in applicable specifications.

3.3 Firing pin assembly load. The firing pin assembly shall not chip, crack or break when subjected to a minimum (min.) tension load of 70 pounds in the test specified in 4.4.1.

3.4 Spring embrittlement. The interrupter spring and tension spring shall not break when extended to the min. point of permanent distortion in the test specified in 4.4.2.

3.5 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substan-

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tiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection.

4.2.1 *Submission.* Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 25 assemblies and 25 complete sets of parts (i.e., 25 each of every component and every subassembly which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 *Inspections to be performed.* Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all of the requirements of the applicable drawings. In addition, for those components requiring a protective finish in accordance with drawing requirements, corrosion tests may be performed in order to determine acceptability of finishes applied.

4.2.3 *Rejection.* If any assembly, subassembly or component part fails to comply with the requirements for the inspections performed in accordance with 4.2.2 the initial production quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions for production.

4.3.1 *Lot formation.* A lot shall consist of inert components produced by one manufacturer in one unchanged process, in accord-

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ance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with Standard MIL–STD–105 (or Standard MIL–STD–1235, when applicable).

4.3.2 *Examination.* Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, support (see dwg. 7549057 covering a detail of dwg. 7549044).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Crack	Visual	01001
Minor:	AQL 2.5 percent		
201.	Outside diameter of flange, min.	Gage	01002
202.	Depth to the bottom	Gage	01003
203.	Outside diameter under flange, min.	Gage	01004
204.	Outside diameter, maximum (max.)	Gage	01005
205.	Diameter of hole, max.	Gage	01006
206.	Evidence of poor workmanship (see 3.5)	Visual	01007

4.3.2.2 Screw, detonator retaining (see dwg. 7549055 or 7549056).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Pitch diameter of thread, min.	Gage	02001
102.	Major diameter of thread, min.	Gage	02002
103.	Slot missing (where applicable)	Visual	02003
Minor:	AQL 4.00 percent		
201.	Width of hexagonal opening (when applicable)	Gage	02004
202.	Total length	Gage	02005
203.	Diameter of hole (when applicable)	Gage	02006
204.	Pitch diameter of thread, max.	Gage	02007
205.	Width of the slot (when applicable)	Gage	02008
206.	Distance from top to the thread (when applicable)	Gage	02009
207.	Evidence of poor workmanship (see 3.5)	Visual	02010
208.	Thread damaged	Visual	02011

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4.3.2.3 Firing pin assembly (see dwg. 7549058 covering a detail of dwg. 7549044).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Length from point to flange	Gage	03001
102.	Total length	Gage	03002
103.	Thickness of flange	Gage	03003
104.	Diameter and profile of pin point	Gage	03004
105.	Pin loose	Manual	03005
Minor:	AQL 1.50 percent		
201.	Diameter under flange	Gage	03006
202.	Diameter of flange	Gage	03007
203.	Evidence of poor workmanship (see 3.5)	Visual	03008

4.3.2.4 Sleeve, setting (see dwg. 7549127 covering a detail of dwg. 7549113).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 1.00 percent		
101.	Location of spring cup and spring cavity	Gage	04001
102.	Depth of spring cup cavity	Gage	04002
103.	Depth of spring cavity	Gage	04003
104.	Diameter of spring cup cavity	Gage	04004
105.	Length to shoulder	Gage	04005
106.	Outside diameter	Gage	04006
107.	Spring cup cavity finish improper	Visual	04007
108.	Location of slot	Gage	04008
109.	Protective finish missing or contains bare spot	Visual	04009
Minor:	AQL 4.00 percent		
201.	Total length	Gage	04010
202.	Diameter of spring cavity	Gage	04011
203.	Diameter of neck	Gage	04012
204.	Width of slot	Gage	04013
205.	Depth of slot	Gage	04014
206.	Diameter of spring cup cavity counterbore	Gage	04015
207.	Depth of spring cup cavity counterbore	Gage	04016
208.	Evidence of poor workmanship (see 3.5)	Visual	04017

4.3.2.5 Cup, spring (see dwg. 7549118 covering a detail of dwg. 7549113).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Larger outside diameter, max.	Gage	05001
102.	Outside diameter finish improper	Visual	05002
103.	Hole missing or obstructed	Visual	05003
104.	Depth of spring cavity	Gage	05004
Minor:	AQL 2.50 percent		
201.	Larger outside diameter, min.	Gage	05005
202.	Diameter of spring cavity	Gage	05006
203.	Length	Gage	05007
204.	Smaller outside diameter	Gage	05008
205.	Evidence of poor workmanship (see 3.5)	Visual	05009
206.	Protective finish missing or contains bare spot	Visual	05010

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4.3.2.6 Spring, interrupter (see dwg. 7549121 covering a detail of dwg. 7549113).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Solid height (max.)	Gage	06001
102.	Outside diameter, max.	Gage	06002
Minor:	AQL 1.50 percent		
201.	Inside diameter, min.	Gage	06003
202.	Ends not closed	Visual	06004
203.	Evidence of poor workmanship (see 3.5)	Visual	06005

4.3.2.7 Interrupter (see dwg. 7549119 covering a detail of dwg. 7549113).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Diameter	Gage	07001
102.	Length	Gage	07002
103.	Finish improper on diameter	Visual	07003
Minor:	AQL 1.00 percent		
201.	Protective coating missing or contains bare spots	Visual	07004
202.	Evidence of poor workmanship (see 3.5)	Visual	07005

4.3.2.8 Retainer, setting sleeve (see dwg. 7549120 covering a detail of dwg. 7549113).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Major diameter of thread, min.	Gage	08001
102.	Pitch diameter of thread, min.	Gage	08002
Minor:	AQL 4.00 percent		
201.	Diameter of thread pilot	Gage	08003
202.	Diameter of hole	Gage	08004
203.	Diameter of hole counterbore	Gage	08005
204.	Depth of hole counterbore	Gage	08006
205.	Length	Gage	08007
206.	Length of thread pilot, max.	Gage	08008
207.	Evidence of poor workmanship (see 3.5)	Visual	08009
208.	Protective coating missing or contains bare spot	Visual	08010

4.3.2.9 Spring, tension (see dwg. 7549122 covering a detail of dwg. 7549113).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Outside diameter, max.	Gage	09001
202.	Inside diameter, min.	Gage	09002
203.	Wire diameter	Gage	09003
204.	Evidence of poor workmanship (see 3.5)	Visual	09004

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4.3.2.10 Tube, flash (see dwg. 7549060).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Pitch diameter of thread	Gage	10001
102.	Major diameter of thread	Gage	10002
103.	Length to end of thread, min.	Gage	10003
104.	Thread damaged	Visual	10004
Minor:	AQL 2.50 percent		
201.	Total length	Gage	10005
202.	Outside diameter	Gage	10006
203.	Length of thread pilot	Gage	10007
204.	Diameter of flash hole for total length, min.	Gage	10008
205.	Diameter of thread pilot	Gage	10009
206.	Evidence of poor workmanship (see 3.5)	Visual	10010

4.3.2.11 Setting sleeve assembly (see dwg. 7549115 covering a detail of dwg. 7549113).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Any component missing	Visual	11001
Minor:	AQL 1.50 percent		
201.	Distance from bottom of spring cup to the end of setting sleeve, max.	Gage	11002
202.	Stake missing or inadequate	Visual	11003
203.	Evidence of poor workmanship (see 3.5)	Visual	11004

4.3.2.12 Ogive (see dwg. 7549126).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Location of setting sleeve hole	Gage	12001
102.	Diameter of setting sleeve hole	Gage	12002
103.	Wall thickness	Gage	12003
104.	Crack, deep scratch or dent	Visual	12004
Minor:	AQL 4.00 percent		
201.	Inside diameter of body seat surface	Gage	12005
202.	Total length	Gage	12006
203.	Outside diameter of head end	Gage	12007
204.	Inside diameter of head end	Gage	12008
205.	Outside diameter at body end	Gage	12009
206.	Location of inside projections	Gage	12010
207.	Distance between inside projections	Gage	12011
208.	Marking misleading or unidentifiable	Visual	12012
209.	Evidence of poor workmanship (see 3.5)	Visual	12013
210.	Letters "SQ" and "Delay" incorrectly positioned	Visual	12014

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4.3.2.13 Body (see dwg. 7549116 covering a detail of dwg. 7549113).

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 1.00 percent		
101.	Pitch diameter of external thread	Gage	13001
102.	Major diameter of external thread, min.	Gage	13002
103.	Minor diameter bottom closing screw thread, max.	Gage	13003
104.	Pitch diameter bottom closing screw thread	Gage	13004
105.	Minor diameter sleeve retainer thread, max.	Gage	13005
106.	Minor diameter flash tube thread, max.	Gage	13006
107.	Pitch diameter flash tube thread	Gage	13007
108.	Concentricity of interrupter cavity with flash hole	Gage	13008
109.	Pitch diameter of sleeve retainer thread	Gage	13009
110.	Length of interrupter cavity protrusion beyond centerline	Gage	13010
111.	Diameter of interrupter cavity	Gage	13011
Minor:	AQL 6.50 percent		
201.	Thickness of boss	Gage	13012
202.	Thickness of flange	Gage	13013
203.	Diameter across wrench slots	Gage	13014
204.	Width of wrench slots, min.	Gage	13015
205.	Location of setting sleeve cavity	Gage	13016
206.	Location of interrupter cavity	Gage	13017
207.	Total length	Gage	13018
208.	Length from base end to bottom of flash tube cavity	Gage	13019
209.	Length from base end to flange	Gage	13020
210.	Length from flange to ogive seat	Gage	13021
211.	Length of external thread undercut, max.	Gage	13022
212.	Length of external thread pilot, max.	Gage	13023
213.	Length of bottom closing screw thread, min.	Gage	13024
214.	Depth of bottom closing screw cavity	Gage	13025
215.	Depth of delay plunger cavity	Gage	13026
216.	Diameter of delay plunger cavity	Gage	13027
217.	Length of flash tube thread pilot, max.	Gage	13028
218.	Diameter of flash tube thread pilot, min.	Gage	13029
219.	Depth of interrupter cavity	Gage	13030
220.	Depth of setting sleeve cavity	Gage	13031
221.	Depth of sleeve retainer thread pilot, max.	Gage	13032
222.	Diameter of setting sleeve cavity	Gage	13033
223.	Diameter of flange, min.	Gage	13034
224.	Angle of flange	Gage	13035
225.	Diameter of external thread undercut, min.	Gage	13036
226.	Diameter of flash hole	Gage	13037
227.	Concentricity of flash hole with delay plunger cavity	Gage	13038
228.	Diameter of body sealing cup groove	Gage	13039
229.	Depth of body sealing cup groove	Gage	13040
230.	Diameter of body sealing cup crimping groove	Gage	13041
231.	Depth of body sealing cup crimping groove	Gage	13042
232.	Diameter of ogive seat	Gage	13043
233.	Finish improper	Visual	13044
234.	Evidence of poor workmanship (see 3.5)	Visual	13045

4.3.2.14 Head (see dwg. 7549061 covering a detail of dwg. 7549044).

Categories	Defects	Method of inspection	Code No.
Critical: None defined.			
Major:	AQL 1.00 percent		
101.	Minor diameter flash tube thread, max.	Gage	14001

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Categories	Defects	Method of inspection	Code No.
102.	Pitch diameter flash tube thread	Gage	14002
103.	Minor diameter detonator retaining screw thread	Gage	14003
104.	Pitch diameter detonator retaining screw thread	Gage	14004
105.	Diameter of firing pin assembly cavity	Gage	14005
Minor:	AQL 4.00 percent		
201.	Depth of flash tube thread, min.	Gage	14006
202.	Depth of flash tube cavity	Gage	14007
203.	Depth to bottom of detonator retaining screw thread, min.	Gage	14008
204.	Diameter of detonator cavity	Gage	14009
205.	Diameter of support cup counterbore, max.	Gage	14010
206.	Length to ogive seat flange from face	Gage	14011
207.	Concentricity of firing pin hole with firing pin assembly cavity	Gage	14012
208.	Major outside diameter of flange, min.	Gage	14013
209.	Depth to bottom of firing pin assembly cavity	Gage	14014
210.	Diameter of ogive seat surface	Gage	14015
211.	Diameter of firing pin hole	Gage	14016
212.	Total length	Gage	14017
213.	Outside diameter at rear end	Gage	14018
214.	Wrench hole or ogive staking hole missing	Visual	14019
215.	Evidence of poor workmanship (see 3.5)	Visual	14020

4.3.2.15 Screw, bottom closing (see dwg. 7549125 covering a detail of dwg. 7549114).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Major diameter of thread	Gage	15001
102.	Pitch diameter of thread	Gage	15002
Minor:	AQL 4.00 percent		
201.	Length of thread plot	Gage	15003
202.	Diameter of thread plot	Gage	15004
203.	Thickness to the wrench hole surface, min.	Gage	15005
204.	Diameter of large counterbore	Gage	15006
205.	Diameter of hole	Gage	15007
206.	Diameter of wrench holes, min.	Gage	15008
207.	Depth of large hole counterbore	Gage	15009
208.	Distance between wrench holes	Gage	15010
209.	Evidence of poor workmanship (see 3.5)	Visual	15011

4.3.2.16 Disc, bottom closing screw (see dwg. 7549124 covering a detail of dwg. 7549114).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Hole, crack or fracture	Visual	16001
Minor:	AQL 1.50 percent		
201.	Diameter	Gage	16002
202.	Thickness	Gage	16003
203.	Evidence of poor workmanship (see 3.5)	Visual	16004

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4.3.2.17 Bottom closing screw assembly (see dwg. 7549114).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Total length, max.	Gage	17001
102.	Component damaged to extent that assembly or function is impaired	Visual	17002
Minor:	AQL 2.50 percent		
201.	Coating on crimp missing or incomplete	Visual	17003
202.	Chamfer missing	Visual	17004
203.	Disc crimp less than 360 degrees	Visual	17005
204.	Wrench hole missing	Visual	17006
205.	Evidence of poor workmanship (see 3.5)	Visual	17007

4.3.2.18 Cup, body sealing (see dwg. 7549117 covering a detail of dwg. 7549113).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical: None defined.			
Major:	AQL 0.40 percent		
101.	Hole, crack or fracture	Visual	18001
Minor:	AQL 2.50 percent		
201.	Depth to bottom surface	Gage	18002
202.	Thickness	Gage	18003
203.	Diameter	Gage	18004
204.	Protective coating missing	Visual	18005
205.	Evidence of poor workmanship (see 3.5)	Visual	18006

4.3.2.19 Body assembly (see dwg. 7549113).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Interrupter spring missing (as determined by tapping assembly lightly against solid object and then sighting through flash hole)	Visual-Manual	19001
2.	Interrupter missing	Visual	19002
3.	Interrupter in armed or partly armed position (does not completely obstruct flash hole)	Visual	19003
4.	Body sealing cup or body closing disc crimp less than 360 degrees	Visual	19004
5.	*Body sealing cup or body closing disc punctured or cracked*	Visual	19005
Major:	AQL 0.40 percent		
101.	Thread damaged	Visual-Gage	19006
102.	Setting sleeve slot incorrectly positioned	Visual	19007
Minor:	AQL 1.5 percent		
201.	Sleeve retainer stake missing or inadequate	Visual	19008
202.	Evidence of poor workmanship (see 3.5)	Visual	19009
203.	Depth of sealing cup (min.)	Gage	19010

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4.3.2.20 Head metal parts assembly (see dwg. 7549044).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	*Length from firing pin point* to detonator cavity (min.)	Gage	20001
Major:	AQL 0.65 percent		
101.	Closing disc damaged	Visual	20002
102.	Firing pin point damaged	Visual	20003
103.	*Firing pin assembly missing*	Visual	20004
Minor:	AQL 2.50 percent		
201.	Length to ogive seat, max.	Gage	20005
202.	Total length, max.	Gage	20006
203.	Wrench hole *missing*	Visual	20007
204.	Closing disc crimp less than 360 degrees	Visual	20008
205.	Evidence of poor workmanship (see 3.5)	Visual	20009

4.3.3 *Testing.*

4.3.3.1 *Firing pin assembly load (see 3.3) —Major defect—Code No. 03009.* The sampling plan shall be in accordance with Standard MIL–STD–105 using Level II and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.1.

4.3.3.2 *Spring embrittlement (see 3.4)—Major defect—Code No. 06006.* Thirty each interrupter springs and tension springs shall be randomly selected for this test. If one or more springs fail to comply with the specified requirements the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.3 *Support cup load (see dwg. 7549057) —Major defect—Code No. 01008.* The sampling plan shall be in accordance with Standard MIL–STD–105 using Level II and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.3.

4.3.3.4 *Arming (see dwg. 7549113)—Major defect—Code No. 19010.* The sampling plan shall be in accordance with Standard MIL–STD–105 using Level II and an AQL of 0.40 percent. This test shall be performed as specified in 4.4.4.

4.3.3.5 *Non-arming (see dwg. 7549113)—Critical defect—Code No. 19011.* This test shall be performed 100 percent in accordance with the procedure specified in 4.4.5. If any body assembly fails to comply with the requirements, it shall be classed defective and removed from the lot.

4.3.3.6 *Torque of setting sleeve (see dwg. 7549113)—Code No. 19012.* The sampling plan shall be in accordance with Standard MIL–STD–105 using Level II and an AQL of 0.40 percent. This test shall be performed as specified in 4.4.6.

4.3.4 *Inspection Equipment.* Index of Inspection Equipment Number IEL–7153449 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

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4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Firing pin assembly load (see 3.3).* The firing pin assembly shall be assembled in equipment that is in accordance with 4.3.4. The required load shall be applied, without impact, and the assembly observed for compliance with the requirements. Assemblies subjected to this test shall be scrapped.

4.4.2 *Spring embrittlement (see 3.4).* The spring shall be pulled to the minimum point of permanent distortion using equipment in accordance with 4.3.4. Springs subjected to this test shall be scrapped.

4.4.3 *Support cup load (see dwg. 7549057).* The support cup shall be placed in equipment that is in accordance with 4.3.4 and the required loads applied without impact. The cup shall be observed for compliance with the requirements. Support cups subjected to this test shall be scrapped.

4.4.4 *Arming test of body assembly.* The unarmed body assembly shall be assembled in the test fixture in a vertical position, with the end containing the flash tube threads up, and be rotated counterclockwise as viewed from the top) at the prescribed arming speed. The spindle shall be of the continuously operating type or it shall start from rest for each test. The rate of acceleration imparted to the body assembly shall be as nearly uniform as practicable for each assembly tested. No assembly, prior to acceptance in the arming test, shall be rotated at a speed greater than that specified. If this test is performed on automatic equipment, it shall be performed prior to the assembly of the body closing disc in the body, after which the closing disc will be crimped in position. The assembly will be classed defective if any part of the interrupter is visible in the flash hole during the arming test.

4.4.5 *Non-arming test of body assembly.* The body assembly shall be tested as specified in 4.4.4 except that rotation shall be at the non-arming speed. If the interrupter does not completely obstruct the flash hole during and following the non-arming test, the assembly shall be classed defective. This test shall be performed subsequent to performance of the non-arming test. If this test is performed on automatic equipment, it shall be performed prior to the assembly of the body closing disc in the body, after which the closing disc will be crimped in position.

4.4.6 *Torque of setting sleeve (see dwg. 7549113).* The torque required to rotate the setting sleeve shall be determined with equipment in accordance with 4.3.4. The body assembly shall be observed for compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packaging, packing and marking of fuze metal parts assemblies shall be in accordance with Specification MIL–P–10025, except appropriate symbol denoting ordnance material not required as part of marking.

5.2 Data cards. Data card information shall be prepared for each lot in accordance with information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify title, number and date of this specification.

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6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the number column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval

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to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be nonreproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of initial production. Instructions as to the location for the evaluation of the initial production sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, Dover, N.J.

6.5 Supersession data. This specification supersedes in part Military Specification MIL–F–12578B (MU).

Custodian:

Army—MU

Preparing activity:

Army—MU

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